



**Politecnico
di Torino**

**Master's Degree:
Energy and Nuclear Engineering**

Nuclear Fusion Reactor Engineering

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1 Introduction

In the following study, a parametric analysis of European DEMO fusion power plant is performed using PROCESS system code designed specifically for a multi-physics reactor problem that couples plasma physics, engineering constraints and cost models. The PROCESS code optimizes a specific variable called Figure of Merit (FoM), in this case it is considered the minimization of the major radius $minmax = 1$ and minimization of the cost of electricity ($minmax = 6$).

2 Case study

Here, the different sweeps performed:

1. Fraction of steel of CS ($nsweep = 66$)
2. Radial thickness of CS ($nsweep = 63$)

3 Steel fraction of CS

In order to perform the following analysis it is used the Figure of Merit = 1, so the Tokamak is in the most compact configuration. The main objective is to understand how the structure reacts when the steel fraction in the Central Solenoid is varied; looking at the hoop stress and the number of cycles until fracture.

3.1 Minimization of the major radius

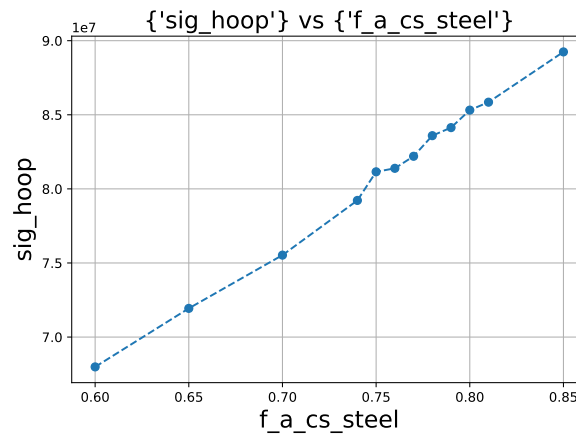


Figure 1: CS hoop stress [Pa] as function of steel fraction

The nearly linear increase of σ_{hoop} in Figure 1 can be understood from the magnetic-pressure balance in a solenoidal coil. For a thin cylindrical winding of radius r and radial thickness

t , the hoop stress can be approximated by

$$\sigma_{\text{hoop}} \simeq \frac{BJr}{t}, \quad (1)$$

where B is the self-field, which can be computed as:

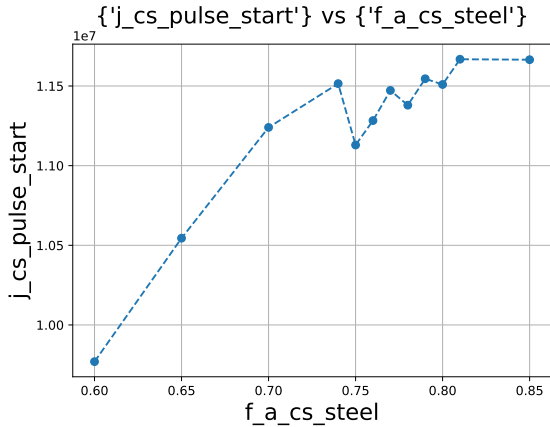
$$B = \mu_0 t J \lambda \quad (2)$$

With μ_0 the vacuum permeability and λ the filling factor, which represents the ratio of the cross-sectional area of superconducting material to the total cross-sectional area of the conductor. Notice that equation 2 refers to the all solenoid, instead of equation 1 to the single conductor.

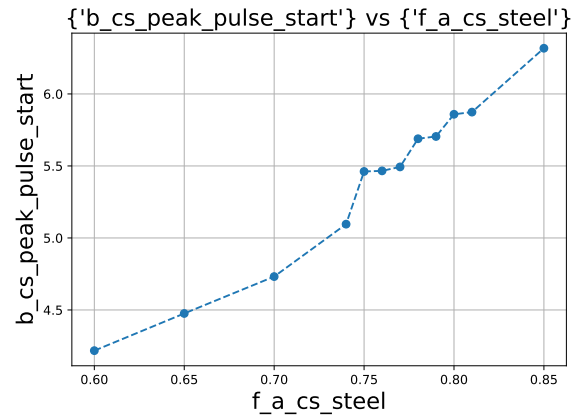
As the steel fraction f_{steel} increases, the available superconducting area decreases.

$$A_{\text{sc}} = (1 - f_{\text{steel}} - f_{\text{void}}) A_{\text{pack}}$$

Hence, with constant current, the current density $J = I/A_{\text{sc}}$ in the superconductor should increase. This is our case; in fact, the figures 2a and 2b show the central solenoid current density and the maximum field in central solenoid both at the start of the pulse (BoF) as a function of the fraction of steel.



(a) Current density [A/m^2] as function of steel fraction



(b) Central solenoid self-field [T] as function of steel fraction

It can be seen that the magnetic flux has more or less the same trend of the σ_{hoop} , however, its dependence on the magnetic pressure should be quadratic ($\sigma_{\text{hoop}} \propto \frac{B^2}{2\mu_0}$), therefore, there must be other parameters that try to reduce the stress, such as the thickness of CS, which increases as expected (figure 3).

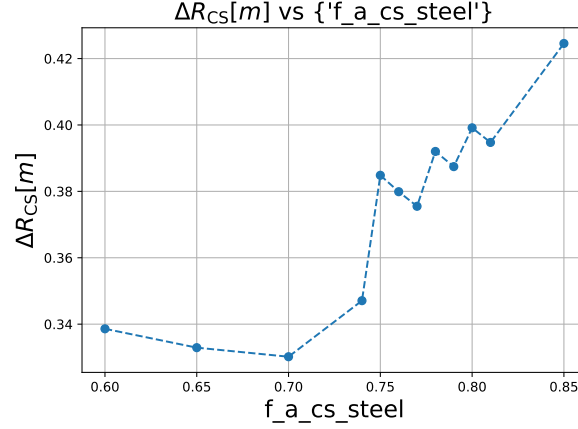


Figure 3: Thickness CS [m] as function of steel fraction

Also the current density increases, however, after 0.75, it looks like to be arrived at saturation. This behavior suggests that PROCESS reaches a point where increasing current no longer brings significant benefit, possibly due to constraints on critical current, temperature margin, or magnetic stability. The magnetic field continues to rise primarily due to geometric effects, such as the expansion of the winding radius.

Over the same sweep window, the allowed number of cycles until the CS fracture evolves as shown in Figure 4, highlighting the trade-off between mechanical stress and fatigue performance.

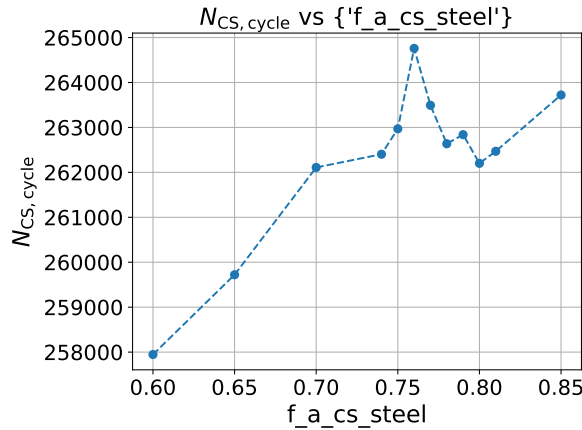


Figure 4: Number of cycle as function of steel fraction

Figure 4 reveals that the admissible fatigue life of the central solenoid rises from $f_{\text{steel}} = 0.60$ to $f_{\text{steel}} \simeq 0.75$, after which it drops slightly. The overall rising trend is probably due to a steel rich winding pack possesses higher fracture toughness and lower Paris law constants,

$$\frac{da}{dN} = C (\Delta K)^m,$$

so crack growth per load cycle is slowed and the component survives more pulses. Beyond $f_{\text{steel}} \approx 0.75$ the marginal gain in fatigue life becomes negligible, whereas both the machine major radius R_{major} and the central solenoid hoop stress continue to increase almost

linearly, undermining compactness and structural margins. Accordingly, a steel fraction of about 0.75 represents a practical upper bound, balancing structural endurance and electromagnetic penalties.

The geometric impact of a winding pack with a higher amount of steel can be appreciated in Figure 5, which tracks the plasma major radius R_{major} returned by PROCESS.

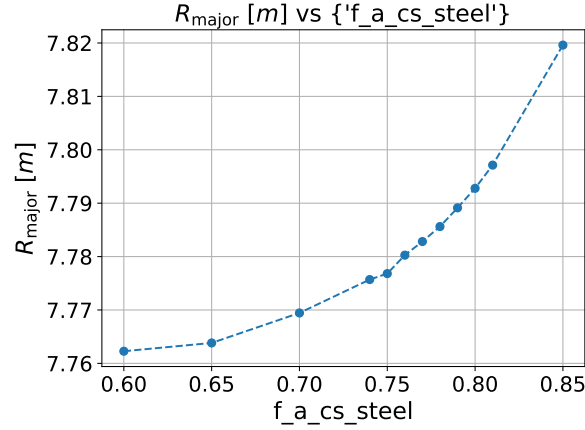


Figure 5: Plasma major radius as function of steel fraction in the CS winding pack

The plasma major radius increases even though the optimizer is configured to minimize it, probably because the structural and spatial constraints introduced by the additional material lead PROCESS to accept a net increase in the overall reactor size, shifting the center of the plasma outward. Despite an increase of only 6 cm, R_{major} defines the overall plant scale, even small changes impact the cryostat volume, TF and PF coil length.

3.2 Minimization of the cost of electricity

In this sweep, the Figure of Merit is set to `minmax = 6`, which minimizes the total cost of electricity production. This objective prioritizes economic performance over compactness or mechanical reserve margins.

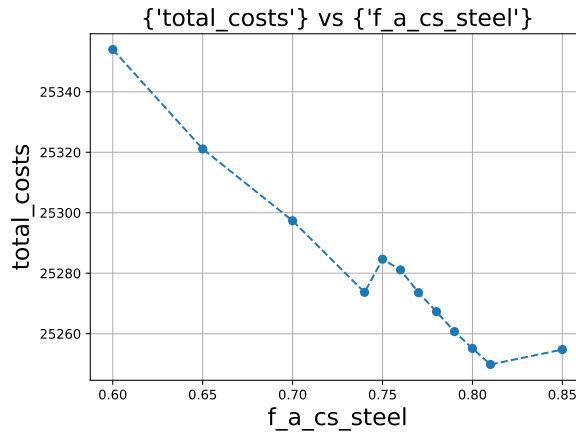


Figure 6: Total plant cost as a function of steel fraction in the CS winding pack

As shown in Figure 6, increasing the steel fraction from 0.60 to 0.85 results in a decreasing of total costs, from approximately 25.35 B\$ to 25.25 B\$.

While this result may seem counterintuitive since a higher steel fraction means using more structural material, it can be explained by the way the "minmax" balances mechanical strength and cost. A winding pack with an higher fraction of steel can handle higher mechanical loads, which allows the design to work with smaller margins and potentially higher current densities. As a result, the volume of superconductor required can be reduced. Even though Nb_3Sn is less expensive than HTS materials, it is still much more costly than steel, so reducing its use helps lower the total cost. Furthermore, adding more steel improves the fatigue resistance of the coil, which can reduce maintenance needs and long term operational costs. Overall, these effects combine could produce a net reduction in total plant cost, but the overnight cost takes into account a lot of other variables.

4 Central Solenoid thickness

In this section a parametric sweep is performed on the radial thickness of the Central Solenoid, with the FoM set on 1. The aim is to investigate the machine compactness and operational performance, when increasing the mechanical robustness of the CS.

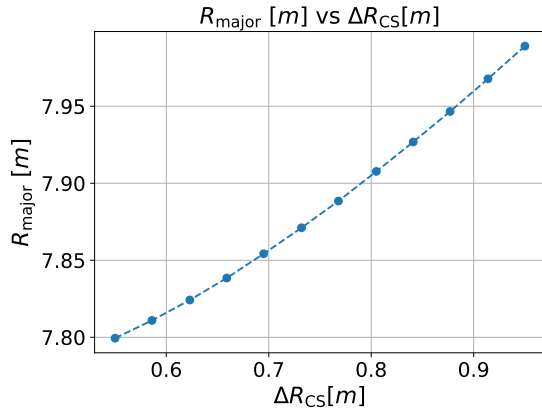


Figure 7: Plasma major radius as a function of CS thickness.

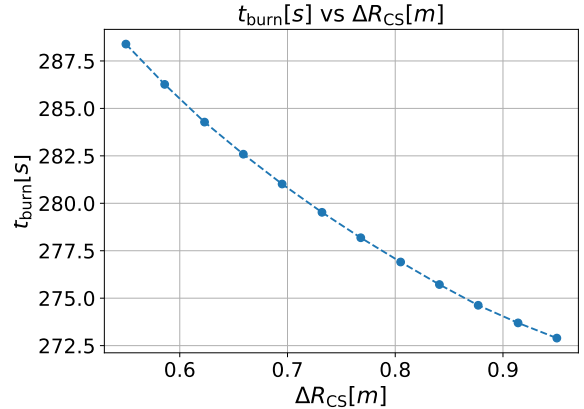


Figure 8: Burn time as a function of CS thickness.

In Figure 7 shows that increasing ΔR_{CS} from 0.6 to 0.93 m causes R_{major} to grow nearly linearly from 7.80 m to 7.96 m. As expected, thicker solenoid increases the inboard radial build and shifts the plasma outward. Although the FoM seeks to minimize R_{major} , this shift is unavoidable due to constraints on shielding and coil geometry.

As Figure 8 shows, the burn time has an inverse correlation, while the CS becomes thicker, it drops from 287.5 s to 272.5 s. This is counterintuitive, as a larger CS is typically expected to offer a greater flux swing. However, the FoM reduces the space available for TF coils and inner shielding, or raises mechanical stresses that limit allowable pulse length. The result is a trade off where mechanical improvements lead to performance penalties.

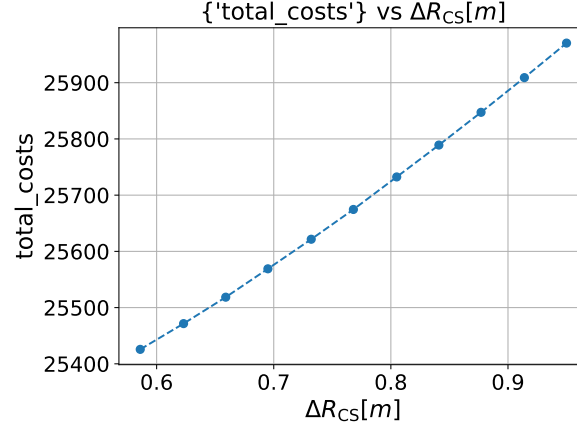


Figure 9: Total plant cost as a function of CS thickness

Figure 9 shows that the total plant cost increases linearly with the CS thickness, ranging from 25.4 to 25.9 B\$. This behavior is consistent with the trends previously observed; a thicker solenoid leads to a larger reactor, both in terms of R_{major} and structural volume. The increased inboard build not only requires more structural and superconducting material, but also enlarges other components, as the cryostat, the vacuum vessel and the magnetic systems. These geometric and material penalties directly affect capital and fabrication costs.

5 Conclusion

The parametric analysis performed with PROCESS yields a clear view that even a single design change can propagate through the system, affecting multiple coupled parameters. By varying the steel fraction in the winding pack and the radial thickness of the solenoid, two distinct scenarios were analyzed, minimization of the major radius and minimization of the cost of electricity.

In the first case the hoop stress on the winding pack rises but the values (70–90 MPa) remain comfortably below the cryogenic yield strength of steel. The strictly connection with the magnetic field was evident, with a following focus in the role of the CS thickness in this quantity but also in the capability to improve robustness, leading to a slightly increase of the number of cycles.

In the second case, by performing a sweep on the CS thickness while minimizing the major radius, it was observed that increasing the solenoid thickness leads to a larger major radius despite the Figure of Merit remaining equal to one and to a shorter pulse length. Although a thicker CS may offer improved robustness, it comes at a higher price.